Contents

Introduction

1
Mathematical modelling

Maths is used to make sense of the world around us, and help us to make decisions, in our personal lives and in the worlds of business, politics, science, and so on. This could involve, for example:

- Creating and solving equations.
- Plotting graphs.
- Making calculations.
- Using computer software.

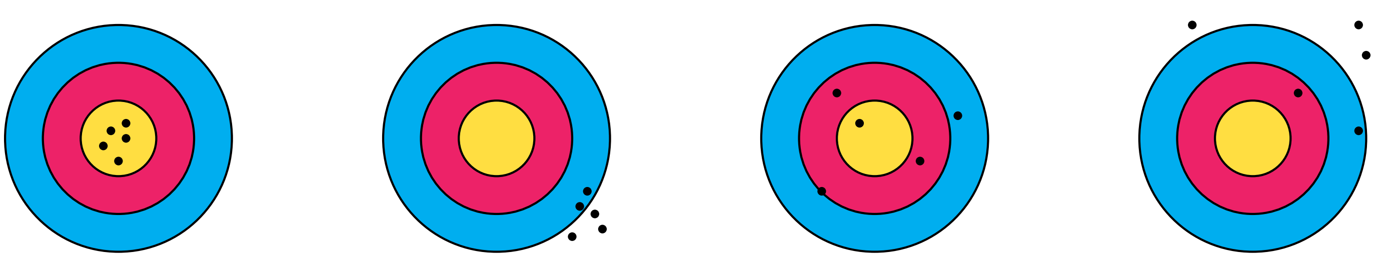
Each of these things could be thought of as creating a **mathematical model**. This chapter will introduce and explore a range of different kinds of model that may be useful, and when they may be encountered in the real world.

> **Quote**
>
> *"All models are wrong, but some are useful."*
> George Box, Statistician

The above quote highlights that mathematical models don't always need to be able to produce 'the *right* answer', and in many situations such a thing is not even possible. Instead, the goal is often a simply a *useful* answer. Since we are entering a world of often *not knowing the true answer*, it is helpful to be learn some related vocabulary:

- **Accuracy** relates to how much a measurement or process leads towards the *true value*.
- **Precision** relates to how close repeated measurements or processes are to *each other*.

The results of four different archers aiming for the centre of a target should help illustrate the difference:



Accurate ✓	Accurate ✗	Accurate ✓	Accurate ✗
Precise ✓	Precise ✓	Precise ✗	Precise ✗

1.1 Systematic Estimation

The 20th century physicist Enrico Fermi was known for performing 'back of an envelope' calculations which often gave surprisingly reliable and *accurate* results. Calculations of this type are therefore sometimes called *'Fermi estimates'*. The goal of these *systematic estimates* is to quickly get a sense of the *scale* of numerical answers to complex or even impossible questions such as:

Questions

- *How many times might a teacher take a class register in their career?*
- *How much on average does a large supermarket spend on staff wages each week?*
- *How many miles are driven by cars on Scottish roads in a month?*

These calculations might involve using a mixture of *approximated*, *estimated* and *exact* numbers. For example:

- There are **exactly** 60 seconds in one minute.
- There are **approximately** 30 days on average in a month.
- We might **estimate** the mass of a typical family car to be 1000 kilograms.

It is important to communicate clearly any **assumptions** made in reaching your answer, such as the *estimation* above.

Example 1.1A

Problem:
A secondary school in a city centre in Scotland has approximately 800 pupils.
Some buy their lunch in the dining hall and some buy their lunch from shops or bring their lunch from home.
Estimate the total amount spent on lunches in the school's dining hall per month.

Solution:
Assume that:

- Half of the pupils in the school buy their lunch from the dining hall.
- On average £4 is spent on lunch each day by those pupils.

$$400 \times 4 \times 20 = 32000 \approx \text{£}30{,}000$$

You may have estimated a different proportion of pupils that use the dining hall, and a different average spend. This is okay, within reason. It is expected your final answer will vary based on your estimates of the different *variables*.

Example 1.1B

Problem:
Estimate the total volume of water that a typical person drinks during their childhood.

Solution:
Assume that:

- 'Childhood' lasts 18 years (from 0 to 18).
- On average a child might drink 1 litre of water a day.

$$1 \times 365 \times 18 = 6570 \approx 6600 \text{ litres}$$

Exercise 1.1

1.2 Modelling Relationships using Graphs

We may wish to explore the *relationships* between two or more variables, such as when considering:

Questions

- How will the **money** in my savings account grow over time if I make regular payments?
- What might a car's **braking distance** by when travelling at different speeds?
- How might the distance travelled in a taxi be related to the **fare**?

We should aim to identify where possible for any relationship:

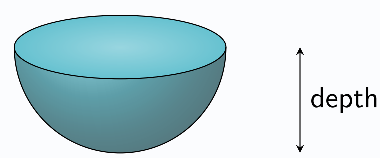
- The independent variable - commonly either *time* or one which we control.
- The **depdendent** variable - which changes in **response** to the other.

Creating graph of a possible model is a good starting point in understanding any relationship. The *independent* variable should be placed along the x-axis and the *response* variable placed on the y-axis.

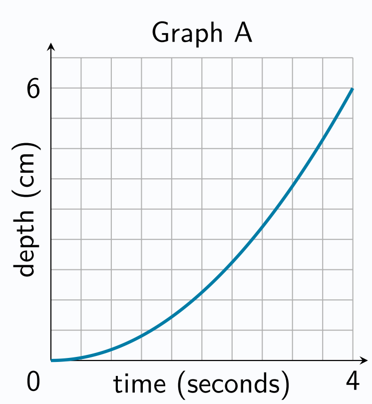
You may have to *sketch* a possible graph of a relationship, or *select from a choice of graphs*, giving an explanation.

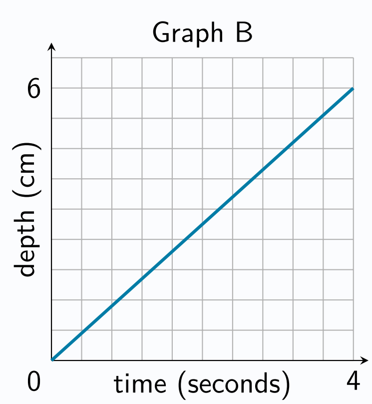
Example 1.2

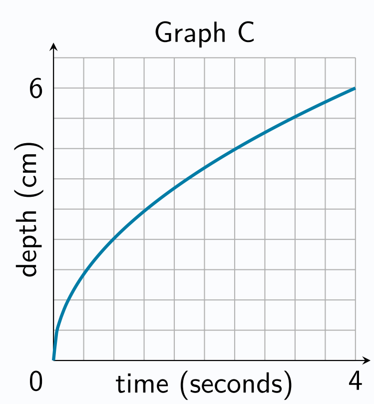
Problem:
Bob pours water into an empty cup at a constant rate.
The interior of the cup is in the shape of a hemisphere, as shown below.



State which of graphs A, B or C below might be best suited to model the relationship between the time since Bob began pouring the water and the depth of water in the cup, explaining your answer.







Solution:
Graph C, since the depth will increase more quickly at the start, then slow down towards the end.

Exercise 1.2

1.3 Linear Relationships

Some types of relationship are commonly encountered, and the rest of this chapter will study these in more detail. One of the simplest relationships between two variables is a *linear relationship*, which on a graph is a *straight line*.

A *positive* linear relationship exists between the price of a taxi ride (P) and the distance travelled (D).	A *negative* linear relationship exists between a candle's height (h) and the time it spends burning (t).

For an equation of the form $y = a + bx$ describing a linear relationship between x and y, the value of a tells us the *starting value* of y (when $x = 0$), and the value of b tells us how much y changes for each 'step' in the x direction.

Exercise 1.3

1.4 Quadratic Relationships

Exercise 1.4

1.5 Exponential Relationships

Exercise 1.5

Review Exercise

Answers

Chapter 1 Answers - Mathematical Modelling

Exercise 1.1

Exercise 1.2

Exercise 1.3

Exercise 1.4

Exercise 1.5

Chapter 1 Review Exercise